

Crafting Figures and Captions

EPRV RCN Observing Proposal Workshop

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Outline

1. Philosophy:
How to decide what your figures to include
2. Aesthetics:
How to design figures that communicate what you want
3. Integration:
How to create a cohesive narrative between the figures and the text

Philosophy

What types of figures can be included and how to decide which

Planning

Figures alone should capture most of the proposal narrative

- Know explicitly what conclusion should be reached for each figure
- Think about how the figures work together to tell the narrative of the proposal
- Planning/iterating figures can be >50% of the time spent on a proposal
 - Pitfall: editing figures can be more fun than writing, but you do have to do both
- Help the reviewers give you a good review

Figure Type: Impact

Convince reviewers to be excited about your result

- If this proposal is successful, here is what we will learn
- Examples
 - “Here be Dragons”: highlight the parameter space your work will expand into
 - Before/after contrasting plots
- Prioritize demonstrating the impact or anticipated change to a field/topic over explaining a topic

Demonstrate (and Label!) Expected Improvement

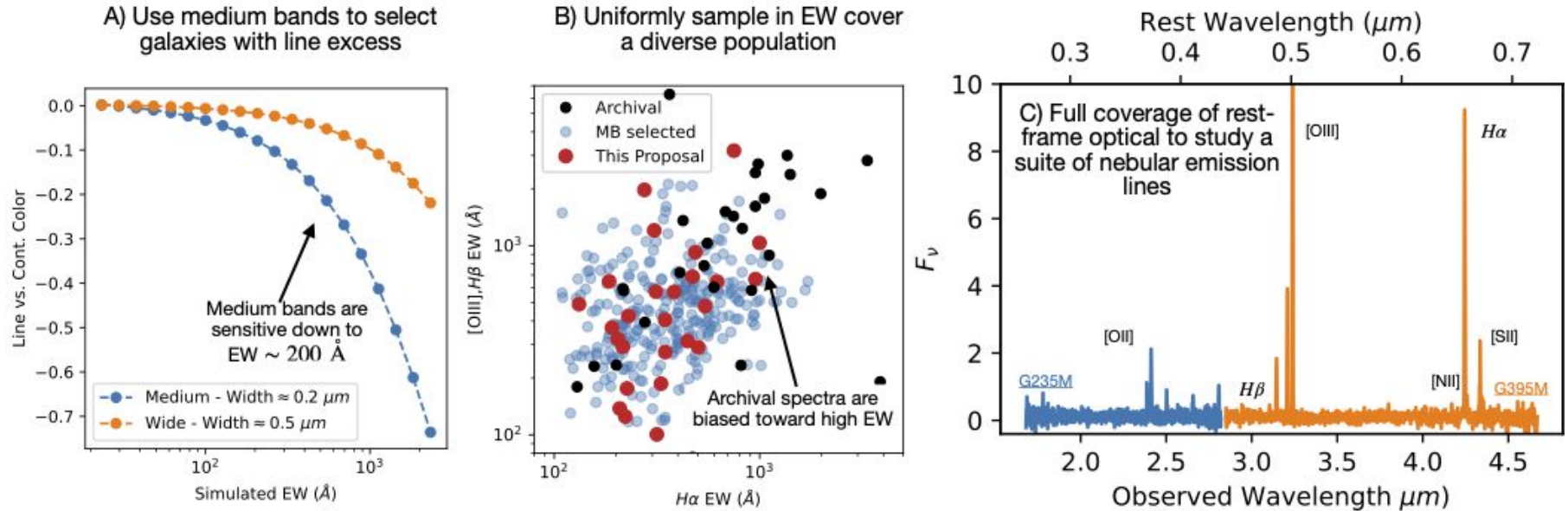
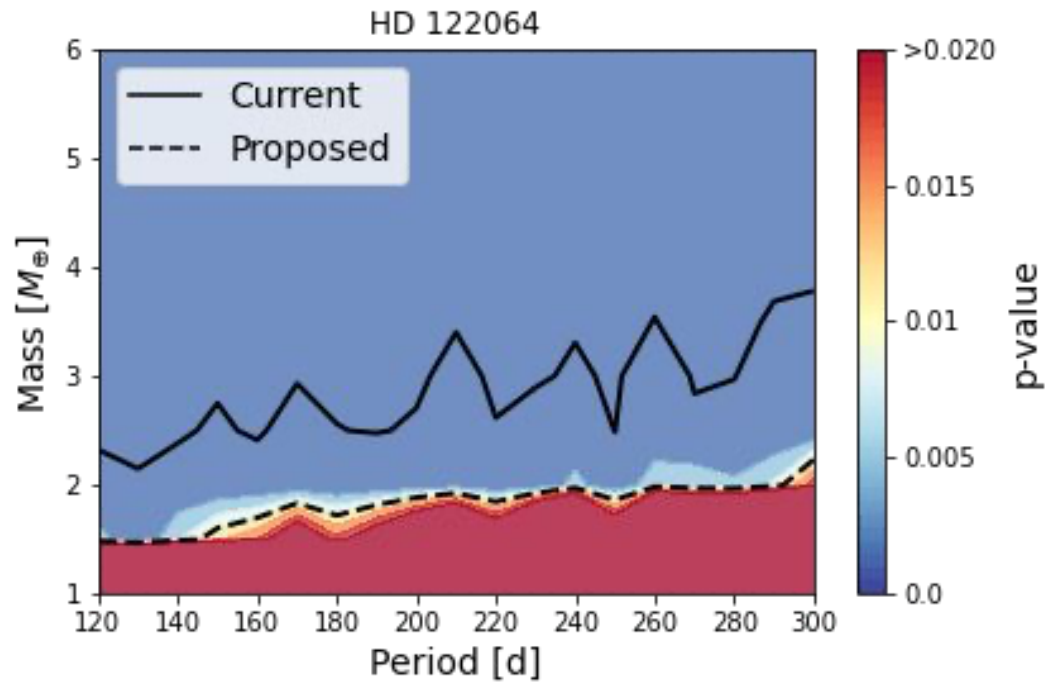


Figure Type: Feasibility

Convince reviewers you're really going to get your result

- It's hard to approve of a proposal if you don't think it will work
- Examples
 - Show that the proposed for data is sufficient for the proposed result
 - Alternatively, demonstrate how much improvement can be expected for different amounts of data (e.g., different time allotments, data quality, etc.)
 - Show that the proposed method/analysis is capable of deriving the result
- Demonstrating a method will work and demonstrating how it works can be a subtle difference

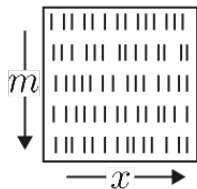
Detectability Curves



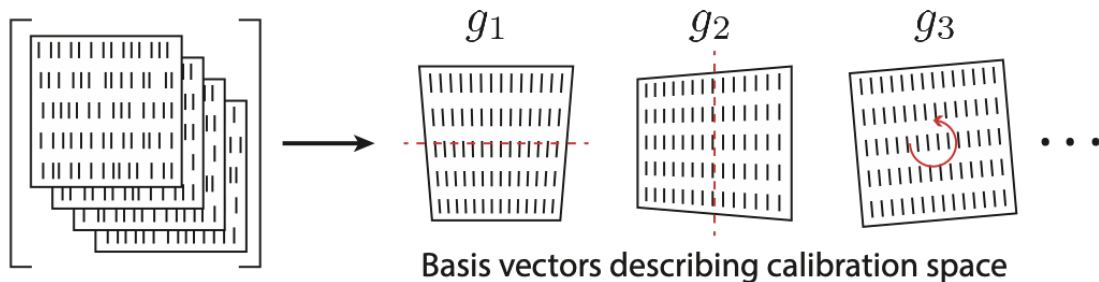
Explanation of Method (bad)

Input

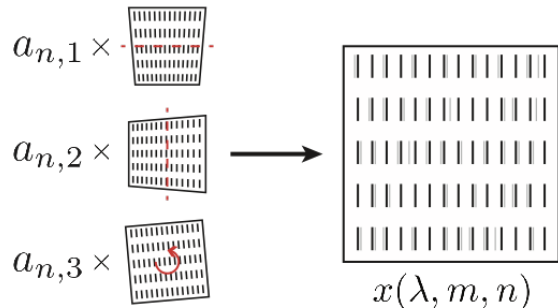
For each exposure, n , a list of calibration line positions, x , with known wavelengths, λ , and echelle orders, m .



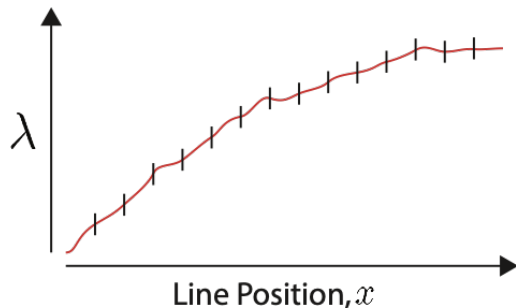
1. Dimensionality Reduction and Denoising



2. Find Calibration State for n



3. Interpolate



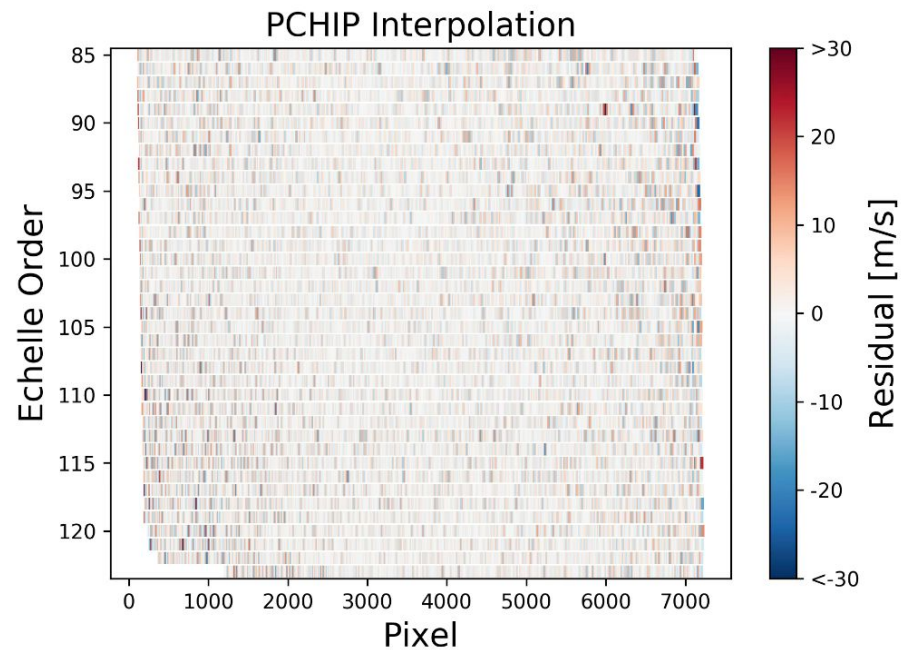
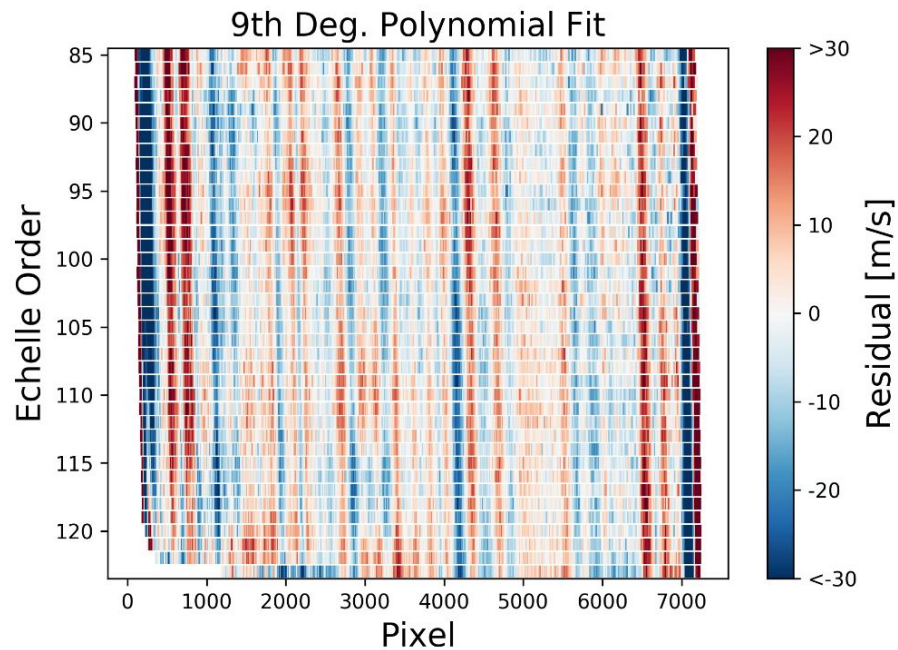
Output

Full de-noised wavelength model

$$x(\lambda, m, n) = g_0(\lambda, m) + \sum_{k=1}^K a_{nk} g_k(\lambda, m)$$

invert to $\lambda(x, m, n)$

Demonstration that Method Works (better)



Aesthetics

Design choices to consider when creating a proposal figure

Design for how your figure is actually viewed

Intentional design choices are noticeable—even subconsciously

- Prefer PDF/vector graphics; use high-DPI PNG for raster images
- Choose the aspect ratio to suit the scientific point
- Design at the final figure size, not just the size on your screen
 - Test at different sizes: screen, print, phone/tablet, e-reader
 - Thumbnail test: is the main point obvious when the figure is small?
- Make the figure interpretable without relying on the caption

Clarify complicated figures

Make the important thing easy to see

- Establish a clear visual hierarchy
- **Lines:** weight, style, opacity, color
- **Markers:** shape, size, fill, color
- **Text:** labels, annotations, statistics
 - Direct-label important curves or regions when possible
 - Nearly universal rule: Your labels can always be bigger!
- Normalize axes/scales across panels when comparison matters
- De-emphasize supporting information

Use color as information

Color should encode meaning, not add decoration

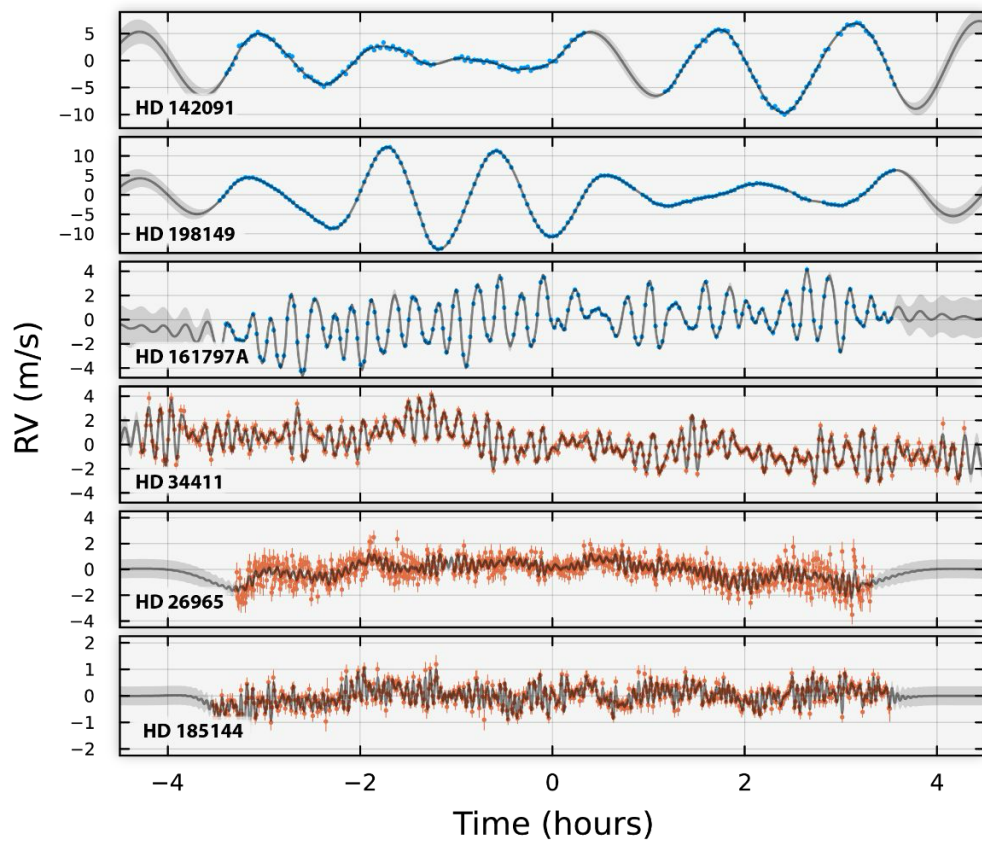
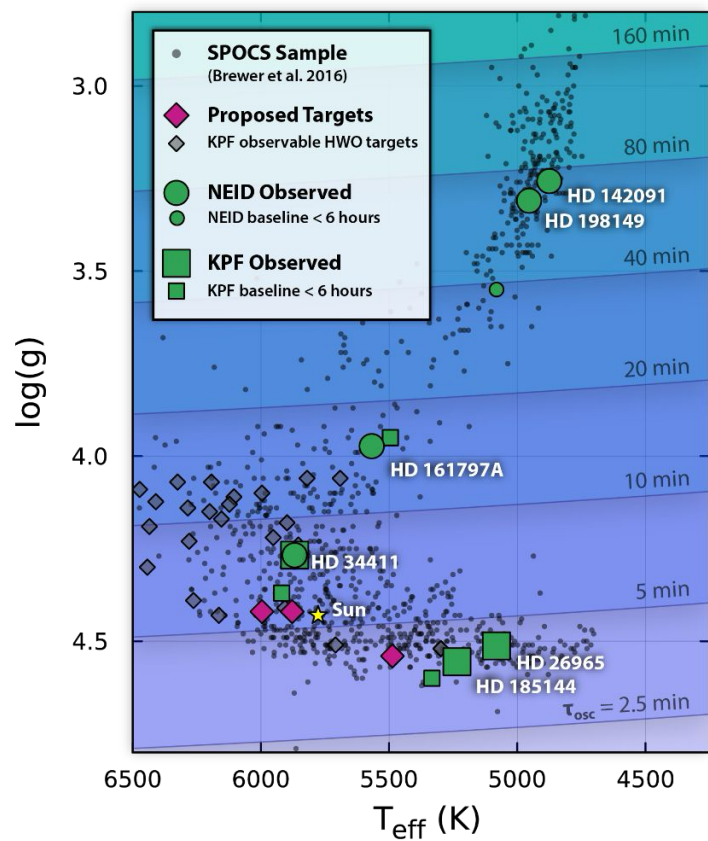
- Match the color palette to the data:
 - Categorical (distinct hues), Sequential (low $\rightarrow$ high), Diverging (meaningful midpoint)
- Use perceptually uniform/color-blind-friendly palettes
- Don't rely on color alone: check figures in grayscale, combine with line style or marker shape
- Build a consistent visual vocabulary across figures:
 - E.g., purple = GP, green = planet; dashed/solid for before/after or observed/predicted
- **Useful tools:** ColorBrewer, seaborn, ColorSchemes.jl, Sim Daltonism

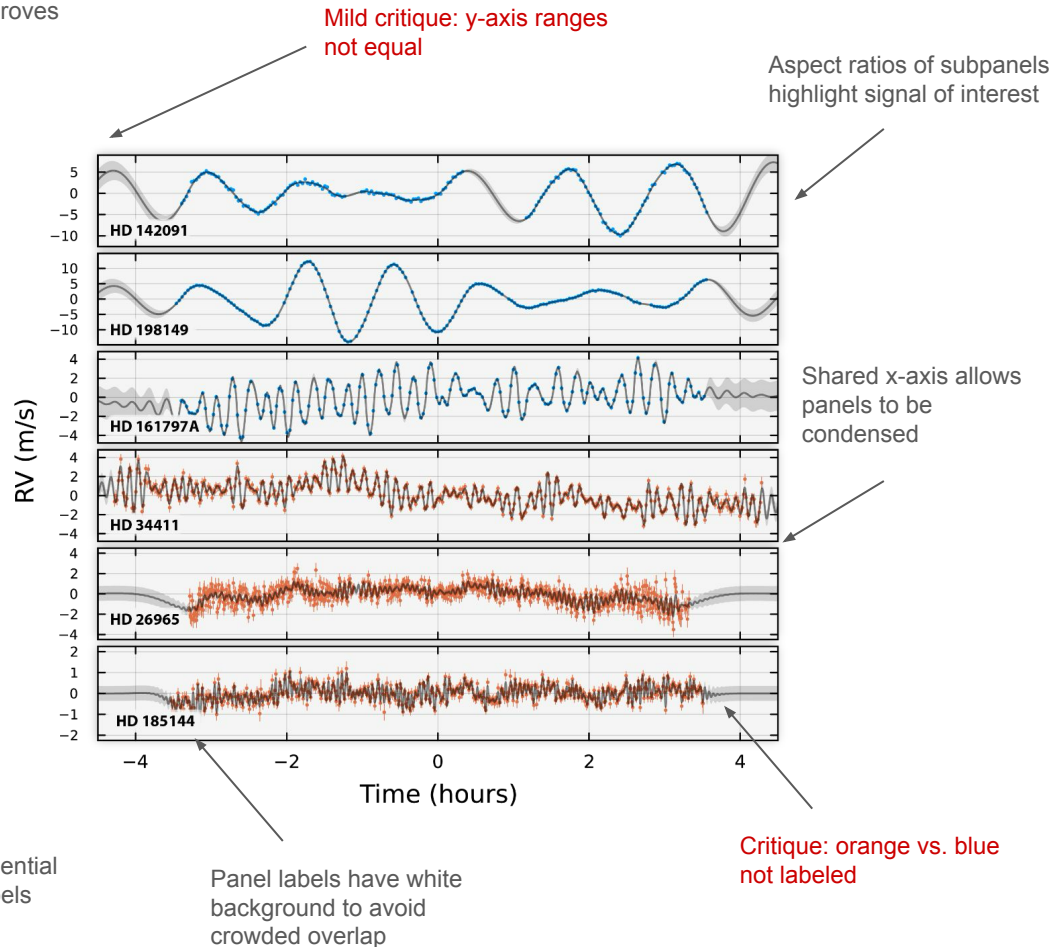
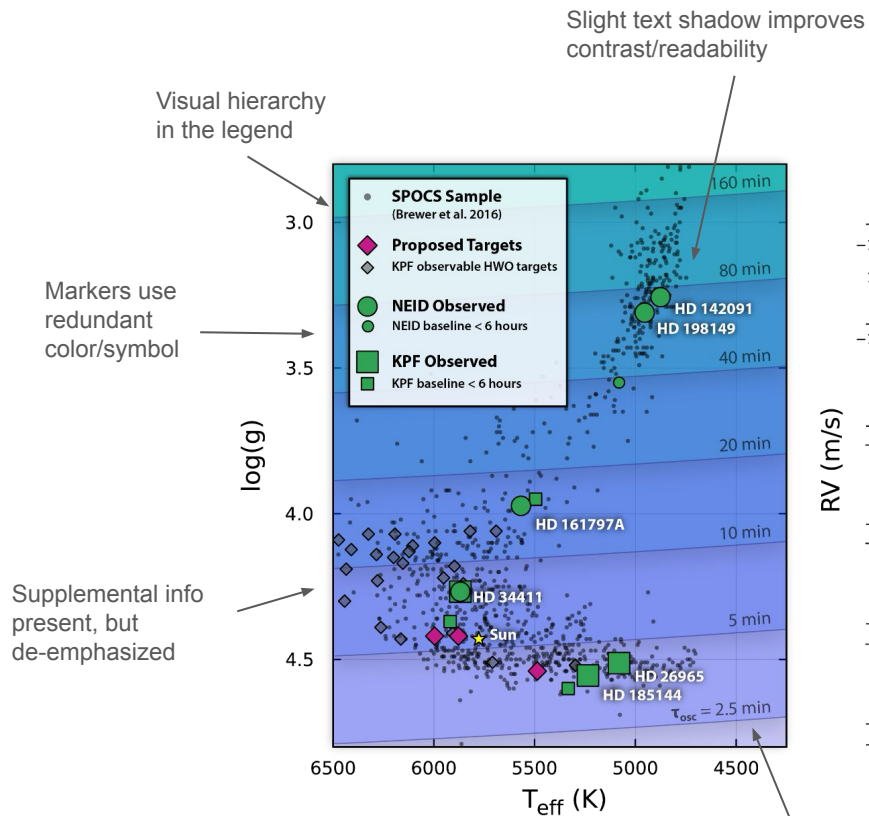
What makes a figure feel “finished”?

A polished figure feels deliberate

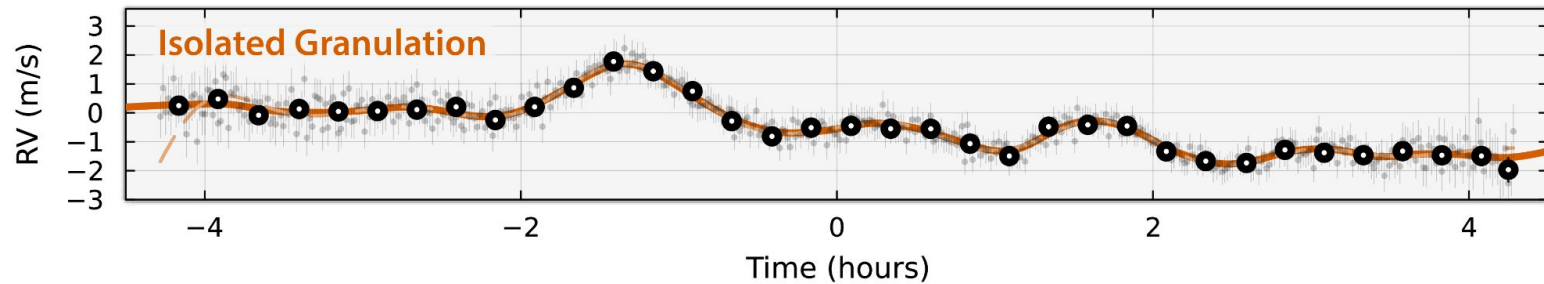
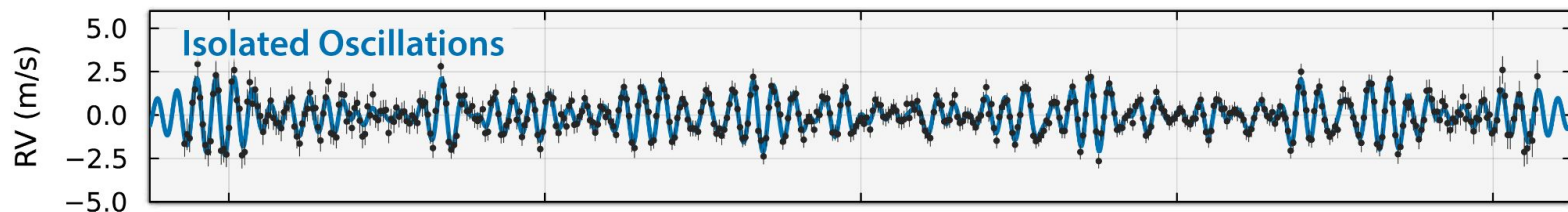
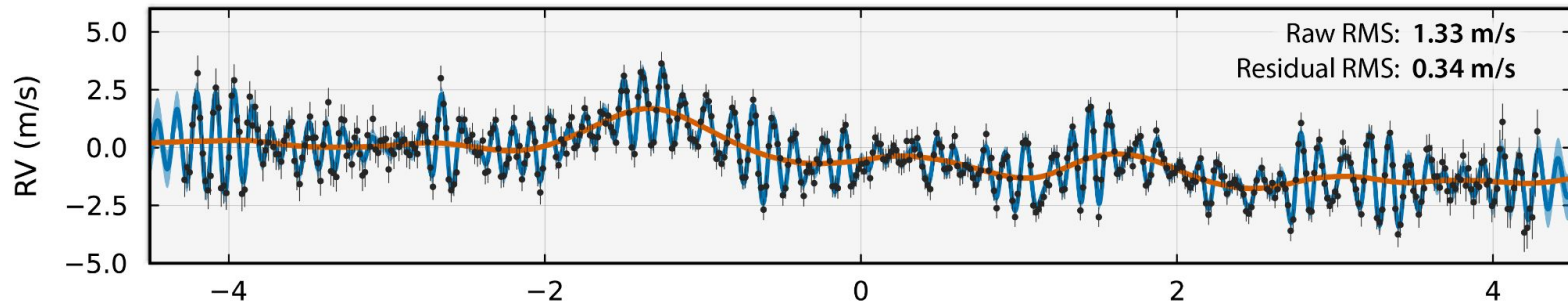
- Use visual hierarchy: **CONCLUSION** → **key data** → supporting content
- Create contrast without making everything harsh or saturated
- Use whitespace deliberately
- Favor direct labels and simple legends
- Shading, boxes, or highlights can help where appropriate
 - Matplotlib path effects
- Keep typography, colors, and styling consistent across the proposal
- Optional finishing workflow: matplotlib → PDF → Illustrator/Inkscape

Learn the rules like a pro, so you can break them like an artist — Picasso

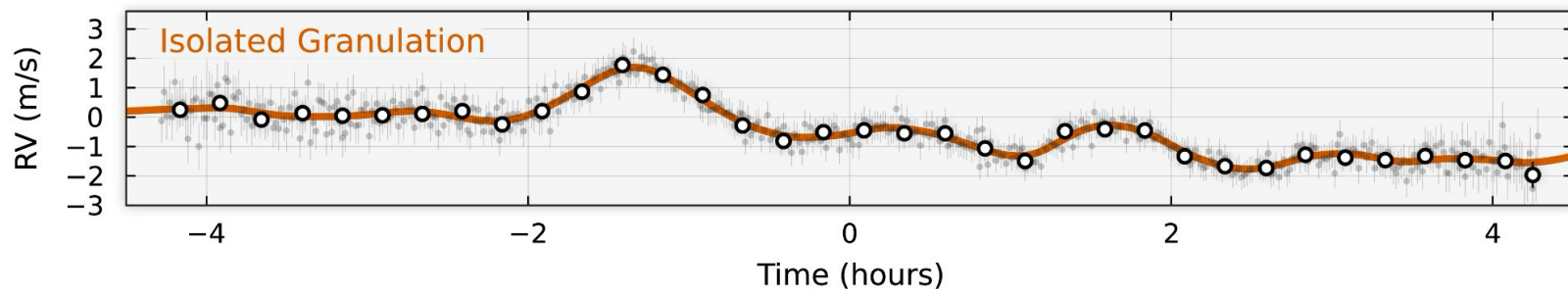
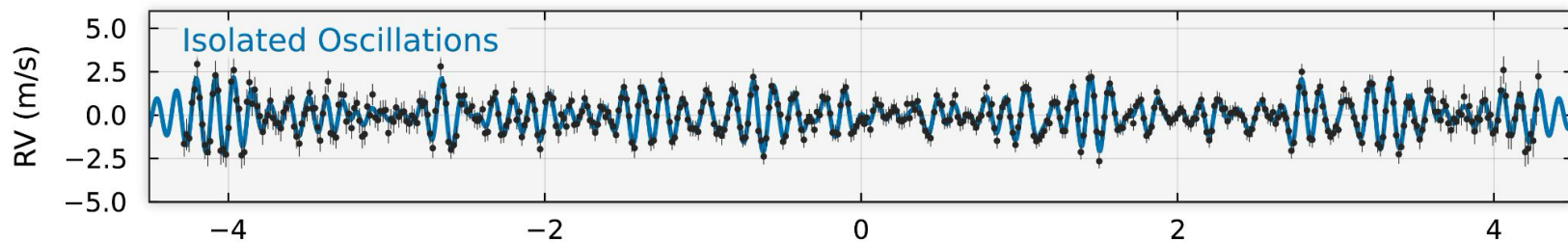
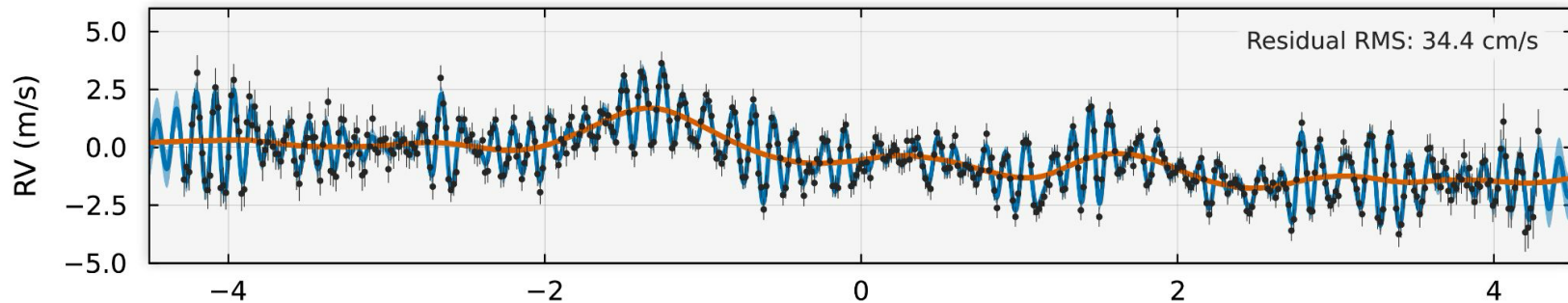


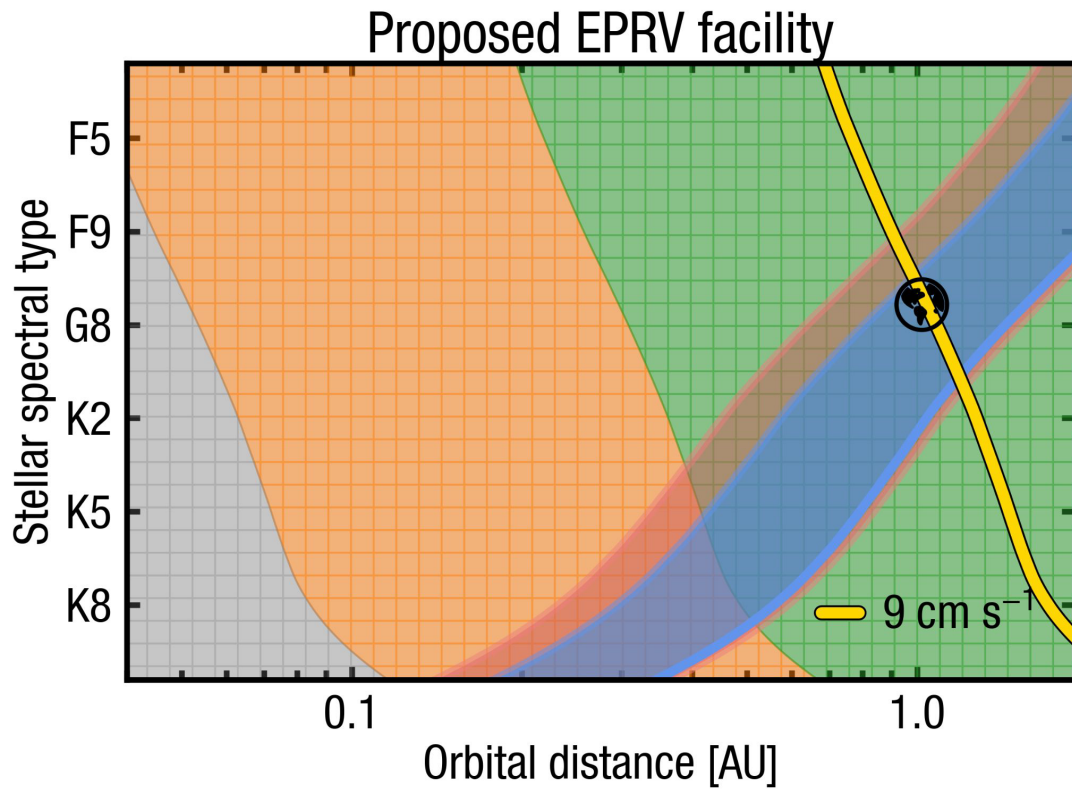


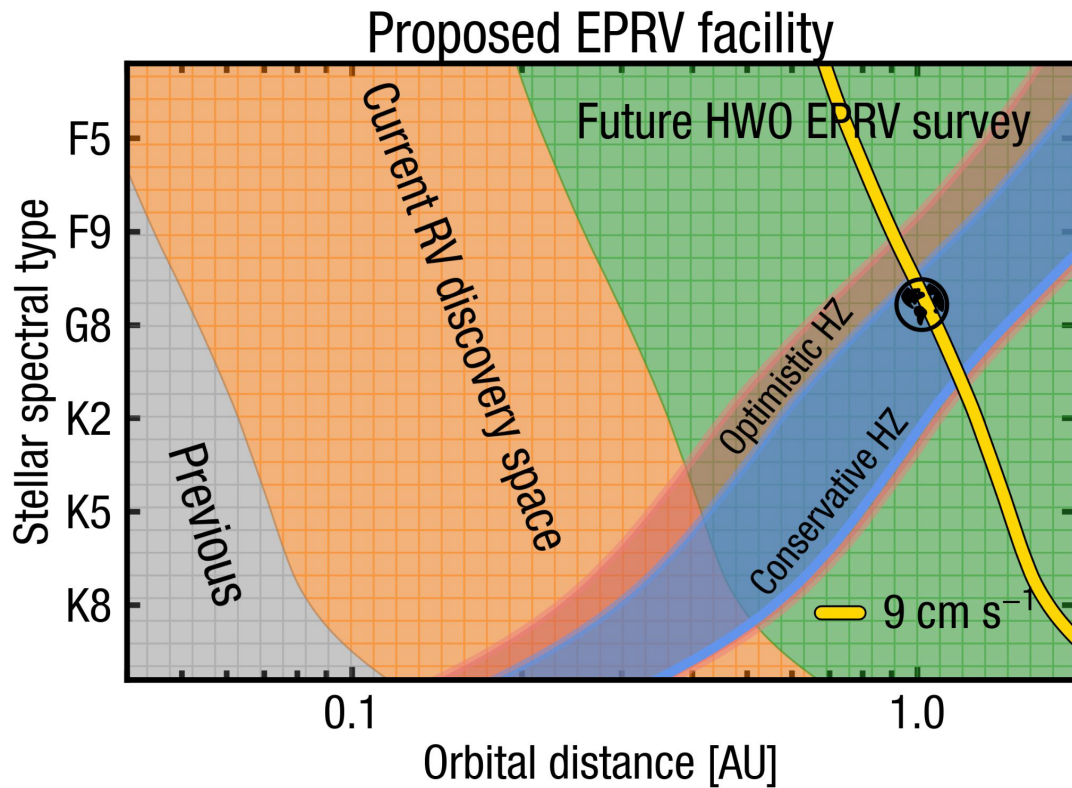
HD 34411 (KPF)



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Integration

How to fit figures and captions in to the proposal narrative

Like figures, all captions should tie to the key points

Captions are the glue that tie the figures to the crux of the proposal

- Captions should be succinct and allow the figures to stand alone. **Some (most) reviewers skim figures/captions first** and/or focus on them during the panel discussion.

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- **Be explicit about the conclusions** you want to be drawn from the figure

What Not to Do

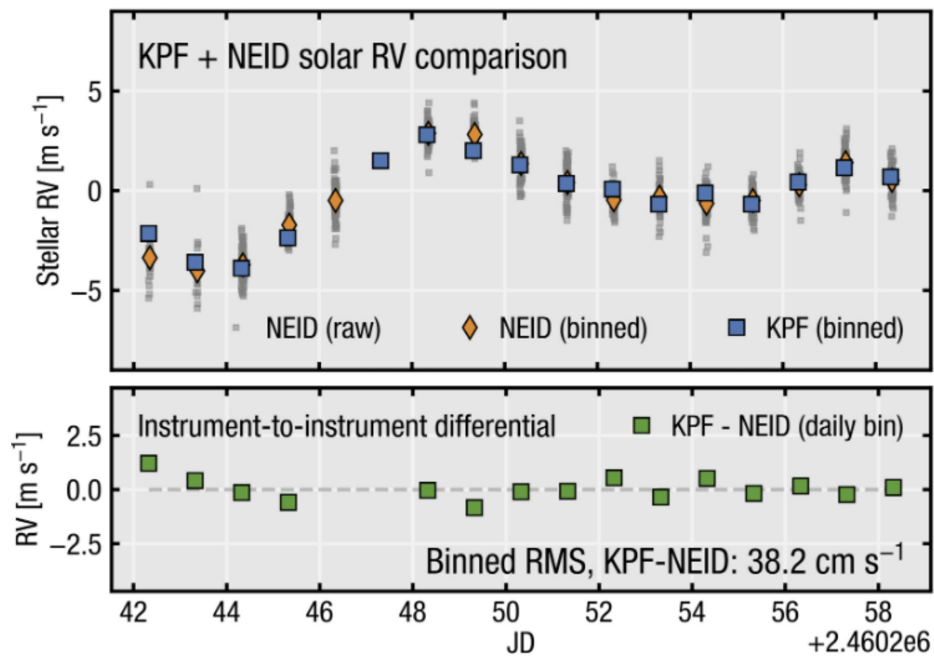


Figure 1: This is the first figure

What Not to Do

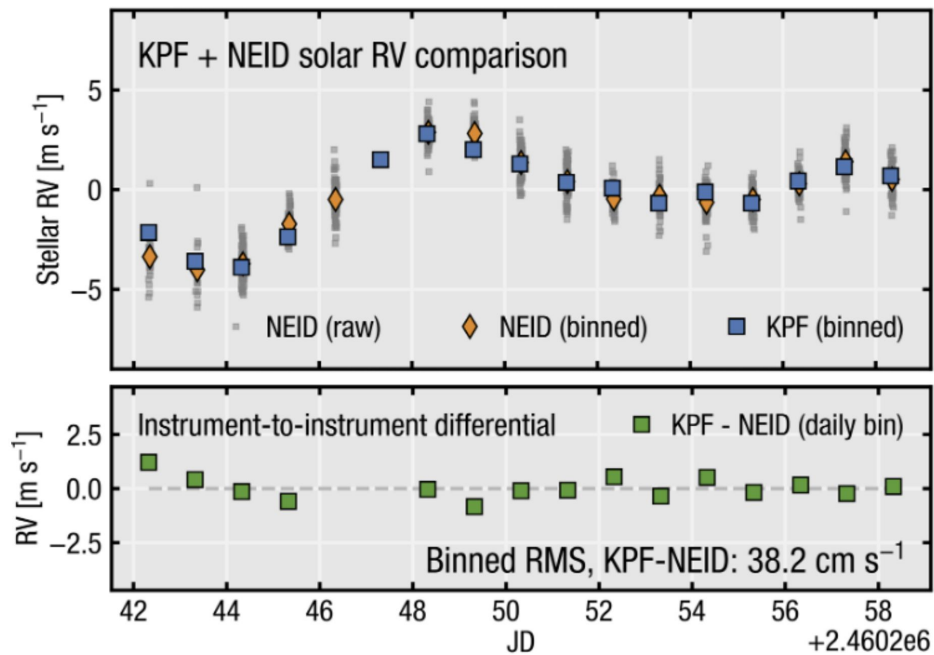


Figure 1: Solar RV time series over two weeks

What Not to Do

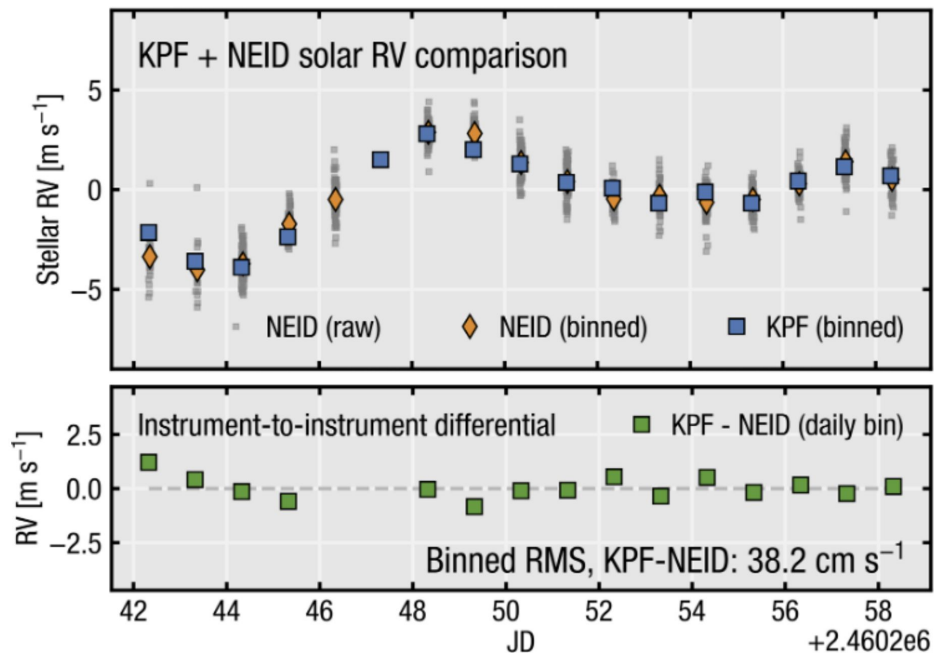


Figure 1: Comparison of solar RV time series from NEID and KPF.

What Not to Do

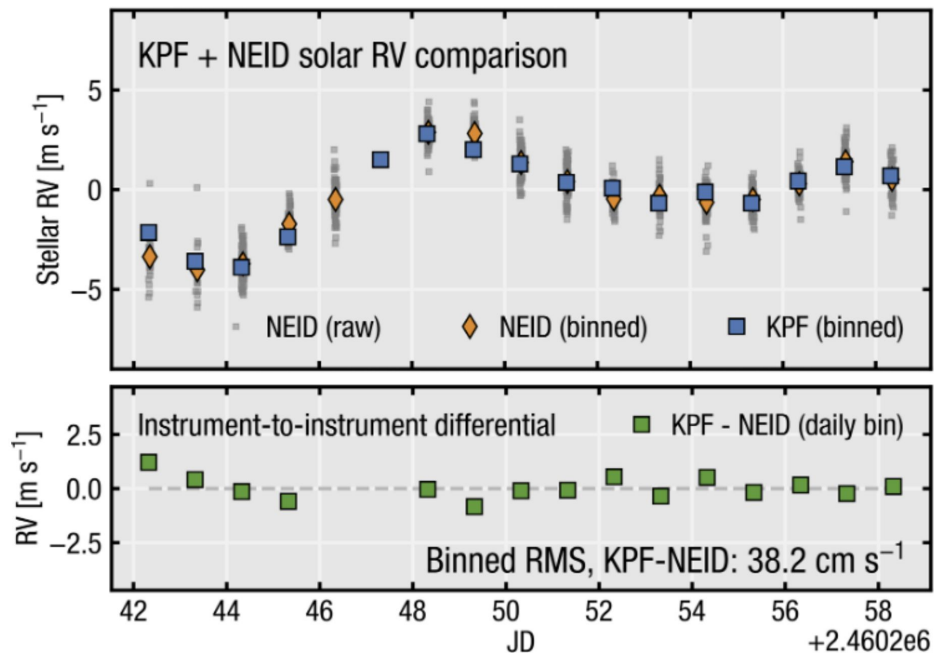
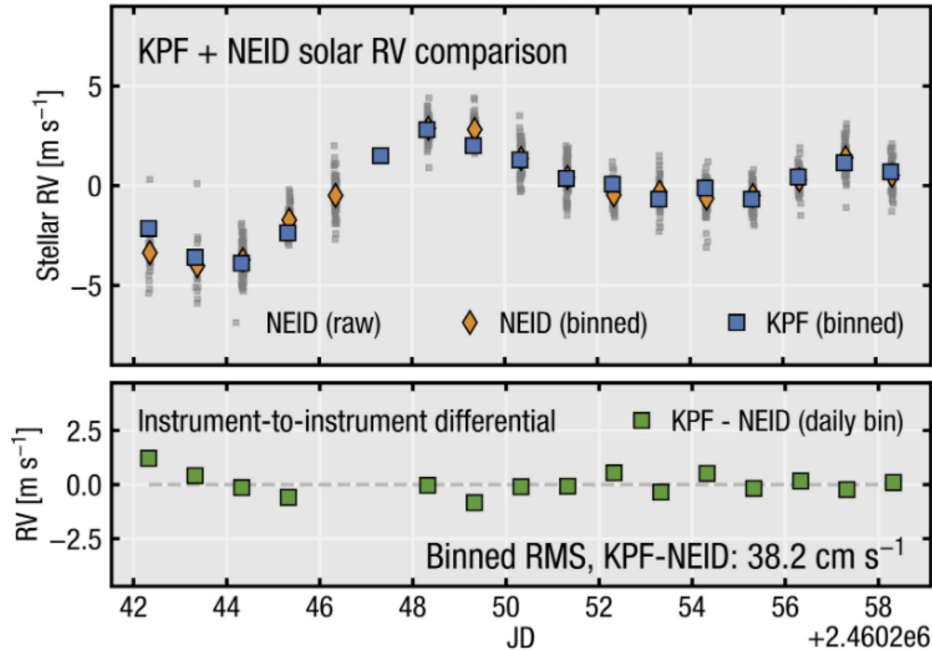


Figure 1: Contemporaneous solar RV time series from KPF and NEID with residuals.

What Not to Do



Describe the figure, why it's exciting, **why it's relevant**

Figure 1: Contemporaneous solar RV time series from KPF (blue squares) and NEID (orange diamonds), highlighting the excellent instrument to instrumental agreement (green lower panel). **Both facilities track the activity signature to $<1 \text{ m/s}$, which is far better than the performance required for our proposed survey ($<5 \text{ m/s}$)**

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go figure